Bayes-Inverse Classification with SmolLM2 and Ensembles for Email Spam Detection

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Abstract

We apply Bayes-inverse classification with the SmolLM2 decoder-only model for spam detection. The method scores label-conditioned prompts using autoregressive log-likelihoods and normalizes them into posterior probabilities. Extending the sequence length and averaging predictions from multiple independently trained models yields large accuracy gains, reaching 97.9%. We also describe the decoder-only architecture, KV-cache implementation, and their role in efficient likelihood computation.

1 Model

1.1 Bayes-inverse scoring

Given an email x and a candidate label $y \in \{\text{ham}, \text{spam}\}$, we create a label-conditioned prompt $P(x, y)$ and compute its log-likelihood under the model:

$$s_y = \log p_\theta(P(x, y)) = \sum_{t=1}^{T} \log p_\theta(w_t \mid w_{1:t-1}).$$

The Bayes-inverse posterior is then:

$$p_\theta(y \mid x) = \frac{\exp(s_y)}{\sum_{y'} \exp(s_{y'})}.$$

This avoids sampling and does not require a classifier head. It only needs the autoregressive likelihood, which SmolLM2 provides directly.

1.2 Fine-tuning objective

When using `method=full_finetune`, we optimize the cross-entropy loss:

$$\mathcal{L} = -\log p_\theta(y_i \mid x_i) = -s_{y_i} + \log \sum_{y'} e^{s_{y'}}.$$

Only one prompt per sample is used during training. Zero-shot and naive-prompting modes freeze the model and skip this objective.

1.3 Our improvements: sequence length and ensembling

Two simple changes gave the most reliable boosts:

Longer sequences. Increasing `max_seq_len` from 256 to 1024 reduces prompt truncation and gives more stable likelihood estimates.

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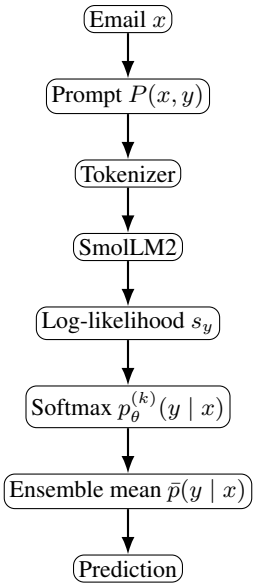

Figure 1: Computation graph for Bayes-inverse scoring and ensembling.

Model ensembling. We train K independent models and average their posteriors:

$$\bar{p}(y \mid x) = \frac{1}{K} \sum_k p_\theta^{(k)}(y \mid x).$$

This reduces variance from small data and random initialization.

1.4 Decoder-only architecture and KV-cache

SmolLM2 is a decoder-only transformer implemented in `model/llama.py`. Tokens flow through stacked `LlamaDecoderLayer` blocks (`model/layers.py`), each containing:

- multi-head causal self-attention (`model/attention.py`),
- RMSNorm (`model/normalization.py`),
- a feed-forward MLP (`model/mlp.py`),
- rotary positional embeddings (`model/positional_encoding.py`).

The attention module enforces a causal mask so the model never sees future tokens. This makes autoregressive likelihoods valid, which is essential for Bayes-inverse scoring.

KV-cache implementation. The KV-cache data structures are in `model/cache.py`. The update logic appears in `model/attention.py`. During inference, keys and values from previous steps are stored and reused. The forward pass concatenates the cached states:

$$K_{\text{new}} = [K_{\text{past}}, K_{\text{current}}], \quad V_{\text{new}} = [V_{\text{past}}, V_{\text{current}}].$$

This reduces computation from $O(T^2)$ to $O(T)$ per generated token once caching begins. The flag `use_cache` in `LlamaModel.forward` enables this behavior.

Why it matters here. Bayes-inverse scoring repeatedly evaluates long prompts, often with shared prefixes. The KV-cache prevents redundant attention computation and makes long-sequence scoring practical.

Drawbacks. The cache grows linearly with sequence length and layer count. Long prompts therefore consume more memory, which explains the GPU pressure observed at large `max_seq_len` values in our ablations.

2 Experiments

2.1 Setup

The released dataset is small, and validation accuracy varies widely across random splits (60–90%). Because of this instability, we train on the full dataset and use the Kaggle public leaderboard as a more stable comparison. All methods use the provided `bayes_inverse_full_finetune` pipeline with adjusted hyperparameters.

2.2 Ablation results

We vary `max_seq_len` and ensemble size K. Each entry averages several runs. Accuracy increases up to 1024 tokens, and larger ensembles steadily help.

Longer sequences help because truncation removes parts of the email that may contain label-relevant information. Keeping more of the text gives the model a fuller likelihood signal.

Ensembling helps because each individual model is noisy on such a small dataset. Averaging their posteriors reduces this noise and produces more stable predictions, with smaller but still positive gains as K grows.

Table 1: Accuracy for different sequence lengths and ensemble sizes.

K	256	512	1024	2048	Avg
1	92.696	94.943	96.488	94.803	94.733
3	91.713	96.769	97.893	96.488	95.716
5	91.853	96.067	97.752	96.769	95.610
7	92.837	96.910	97.331	97.191	96.067
9	93.960	96.910	97.893	97.050	96.453
Avg	92.275	96.172	97.366	96.313	–

Longer sequences help until 1024 tokens. Ensembling provides monotonic but diminishing gains. The best average accuracy is obtained with `max_seq_len = 1024` and an ensemble of $K = 9$ models.

3 Conclusion

Bayes-inverse scoring with SmolLM2 is a simple and effective classifier that outperforms prompting baselines. Longer prompts and ensembling give the largest gains, enabled by the decoder-only architecture and KV-cache. The main limitation is memory growth with sequence length. Future work includes better ensemble selection, synthetic data, and joint label scoring.

References

- DSL-Lab. *CPEN455-Project-2025W1 GitHub Repository*. `https://github.com/DSL-Lab/CPEN455-Project-2025W1`
- CPEN455 Kaggle Competition. `https://www.kaggle.com/competitions/25w1-cpen-455-course-project`